Conservation of Energy

Answer key

Conventions used throughout

• **Energy symbols (from the lessons):** W = work, E_k = kinetic energy ($\frac{1}{2}mv^2$), E_g = gravitational potential energy (mgh), E_e = elastic potential energy ($\frac{1}{2}kx^2$). x is the stretch or compression measured from the spring's unstretched length.

• **Method (from the lesson):** draw the situation, list the kinds of energy at the start and at the end, then write $E_{TOTi} = E_{TOTf}$. With friction or another non-conservative force: $E_{TOTi} - W_F = E_{TOTf}$, where W_F is the size of the work done by that force.

• **Heights:** E_g is relative, so each problem picks its own h = 0 (usually the lowest point of the object).

• **Friction:** $F_f = \mu F_N$, and the work it removes is $W_F = F_f \Delta d$ over the whole distance travelled. The normal force comes from the balanced direction ($F_{net,y} = 0$).

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• **Final answers:** work and energy are scalars (a size and a sign, no direction). Vectors are given as a size with the direction in square brackets.

• g = 9.81 m/s^2. Every step keeps full precision; only final answers are rounded.

CONSERVATION OF ENERGY (A)

Types of energy at the beginning and end

a) identify the type(s) of energy of the object at the beginning and end of each situations

A roller coaster travelling slowly at the top of a hill speeds up as it goes down the hill

A compressed spring fires a dart straight up into the air

A ball is thrown in an arc

A bungee jumper jumps off a tower

E_k needs speed, E_g needs height above the chosen h = 0, and E_e needs a stretched or compressed spring (or cord).

Situation	Beginning	End
Roller coaster	E_g (large, top of the hill) and a little E_k (moving slowly)	Mostly E_k (fast at the bottom); less E_g
Dart fired straight up	E_e in the compressed spring	E_g at the top of its flight (E_k on the way up, zero at the top)
Ball thrown in an arc	E_k as it leaves the hand	E_k again when it comes back down to the same height (E_k + E_g at the top of the arc, where it is still moving sideways)
Bungee jumper	E_g at the top of the tower	E_e in the stretched cord at the lowest point (E_k + less E_g on the way down)

FINAL ANSWER

Roller coaster: E_g + small E_k → mostly E_k

Dart: E_e → E_k → E_g

Ball: E_k → E_k + E_g → E_k

Bungee jumper: E_g → E_k + E_g → E_e

CONSERVATION OF ENERGY (B)

Why the work done by friction is subtracted

b) Explain why the work done by friction is subtracted from the initial energy.

Friction always points against the motion, so the work it does on the object is negative: it takes mechanical energy out of the system and turns it into thermal energy. The system therefore ends with less mechanical energy than it started with, by exactly the amount W_F. Writing $E_{TOTi} - W_F = E_{TOTf}$ uses W_F as a positive size and lets the minus sign do the removing.

FINAL ANSWER

Friction removes energy from the system, so the final energy is the initial energy minus W_F

EXAMPLE 1

Block and spring

a) A 0.50 kg block travels at 5.0 m/s along a frictionless horizontal surface when it encounters a spring of force constant 30.0 N/m. How much will it compress the spring? Include a diagram.

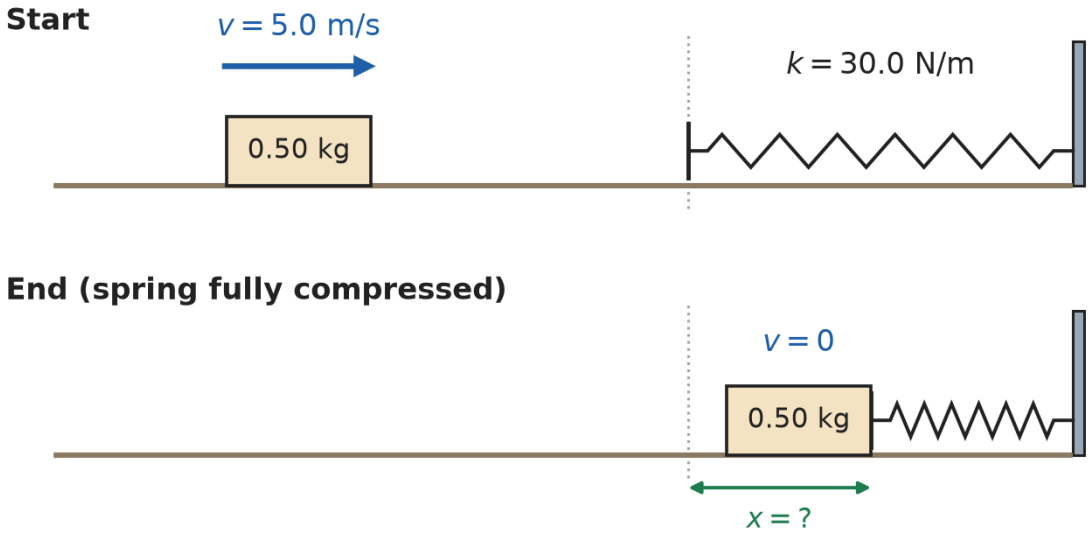
b) If there is 2.0 N of friction between the surface and the block, how much will it compress?

GIVEN

$m = 0.50$ kg $\quad v = 5.0$ m/s $\quad k = 30.0$ N/m

$F_f = 2.0$ N (part b) $\quad x = ?$

The block at the start and at maximum compression



The surface is level, so E_g does not change and is left out.

a) How much will it compress the spring? Include a diagram.

Start: the block is moving and the spring is unstretched, so the energy is all kinetic. End: the block has stopped, so the energy is all elastic. No friction, so energy is conserved.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

Multiply by $\frac{2}{k}$: $x^2 = \frac{mv^2}{k}$

Square root both sides: $x = v\sqrt{\frac{m}{k}}$

Put in the numbers: $x = (5.0)\sqrt{\frac{0.50}{30.0}} = 0.6455 \text{ m}$

b) If there is 2.0 N of friction between the surface and the block, how much will it compress?

Step 1 – Energy equation with friction

Friction acts over the compression distance x, so it removes $W_F = F_f x$.

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies: $\frac{1}{2}mv^2 - F_f x = \frac{1}{2}kx^2$

Move everything to one side: $\frac{1}{2}kx^2 + F_f x - \frac{1}{2}mv^2 = 0$

Put in the numbers: $15.0x^2 + 2.0x - 6.25 = 0$

Step 2 – Solve the quadratic

x appears as x^2 and as x, so use the quadratic formula with a = 15.0, b = 2.0, c = −6.25.

Formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Put in the numbers:

$$x = \frac{-2.0 \pm \sqrt{(2.0)^2 - 4(15.0)(-6.25)}}{2(15.0)} = \frac{-2.0 \pm 19.468}{30.0}$$

$x = 0.5823$ m or $x = -0.7156$ m

A compression cannot be negative, so x = 0.5823 m. It is smaller than in (a) because friction removed some of the energy.

FINAL ANSWER

a) x = 0.65 m = 65 cm

b) x = 0.58 m = 58 cm

Worksheet answer: a) 64 cm
Why the answers differ: Different rounding. (5.0)√(0.50/30.0) = 0.6455 m = 64.55 cm, which rounds to 65 cm. The worksheet cut the number off at 64 cm.

EXAMPLE 2

Roller coaster

A 665 kg roller coaster car starts from rest at the start of the first drop. It rolls down to the bottom of the first drop which is vertically 25 m lower than the top. The actual length of the track from the top to the bottom is 35 m (an angle of 45^0 from the vertical).

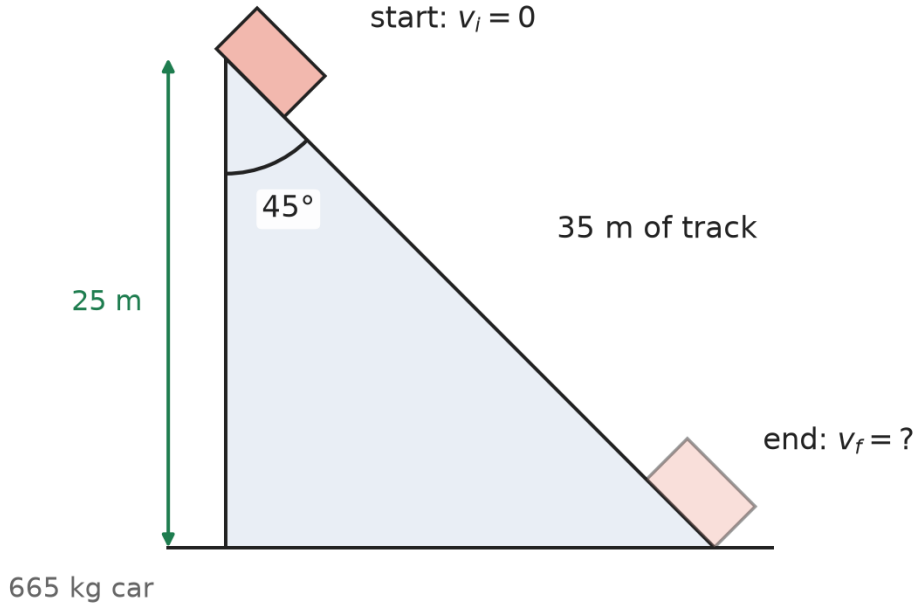
a) Assuming that friction is negligible, determine the speed of the roller coaster when it reaches the bottom of the first drop.

b) If the coefficient of friction is 0.15, what is the speed of the roller coaster car at the bottom of the first drop?

GIVEN

$m = 665$ kg	$v_i = 0$	$h = 25$ m
$\Delta d = 35$ m	$\mu = 0.15$ (part b)	$v_f = ?$

The first drop (h = 0 at the bottom)



Start (top): at rest, so $E_{TOTi} = E_g$. End (bottom, h = 0): $E_{TOTf} = E_k$.

a) Assuming that friction is negligible, determine the speed of the roller coaster when it reaches the bottom of the first drop.

With no friction all of the gravitational potential energy at the top becomes kinetic energy at the bottom. The mass cancels.

Formula: $E_{TOTi} = E_{TOTf}$

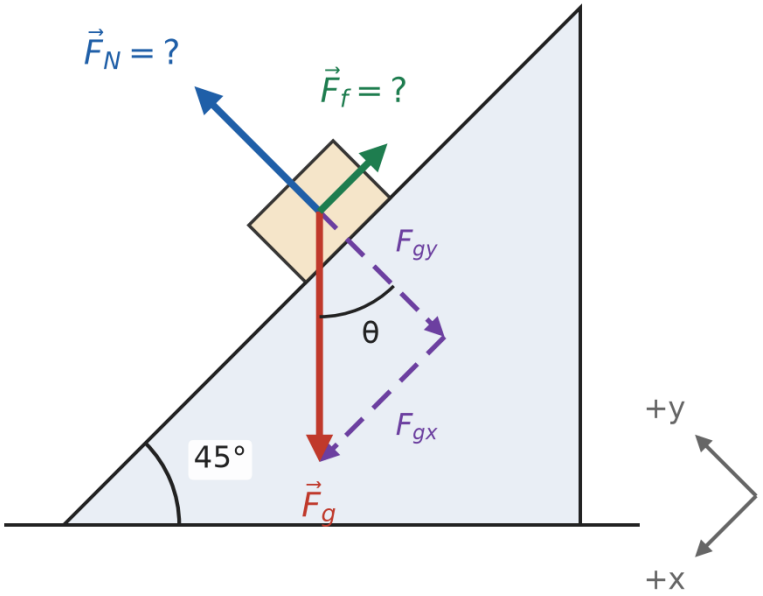
Put in the energies: $mgh = \frac{1}{2}mv_f^2$

Divide by m, multiply by 2, square root: $v_f = \sqrt{2gh}$

Put in the numbers: $v_f = \sqrt{2(9.81)(25)} = 22.147$ m/s

b) If the coefficient of friction is 0.15, what is the speed of the roller coaster car at the bottom of the first drop?

FBD of the car on the track (+x down the track, +y away from the track)



The track is 45° from the vertical, so it is also 45° from the horizontal, and θ = 45° in the gravity triangle.

Step 1 – Normal force (balanced direction)

The car does not accelerate perpendicular to the track. Forces are sizes, with + away from the track:

Balance: $F_{net,y} = 0, \quad \text{so} \quad 0 = F_N - F_{gy}$

Put in $F_{gy} = mg\cos\theta$: $F_N = mg\cos\theta$

Put in the numbers: $F_N = (665)(9.81)\cos 45° = 4612.9 \text{ N}$

Step 2 – Friction and the work it removes

Friction acts along the whole 35 m of track.

Friction: $F_f = \mu F_N = (0.15)(4612.9) = 691.94 \text{ N}$

Work removed: $W_F = F_f \Delta d = (691.94)(35) = 24218 \text{ J}$

Step 3 – Energy equation with friction

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies: $mgh - W_F = \frac{1}{2}mv_f^2$

Multiply by $\frac{2}{m}$ and square root: $v_f = \sqrt{\frac{2(mgh - W_F)}{m}}$

Put in the numbers: $v_f = \sqrt{\frac{2((665)(9.81)(25) - 24218)}{665}}$

$v_f = \sqrt{\frac{2(163091 - 24218)}{665}} = 20.437 \text{ m/s}$

FINAL ANSWER

a) v_f = 22 m/s

b) v_f = 20. m/s

EXAMPLE 3

Applying the law of conservation of energy

A model car of mass 5.0 kg slides down a frictionless ramp onto a horizontal frictionless surface where it encounters a spring k = 4.9 kN/m.

(a) The spring experiences a maximum compression of 22 cm. Determine the height of the initial release point. Include a diagram.

(b) Calculate the speed of the model car when the spring has been compressed 15 cm.

(c) Determine the maximum acceleration of the car after it hits the spring.

GIVEN

$m = 5.0$ kg

$k = 4.9$ kN/m $= 4900$ N/m

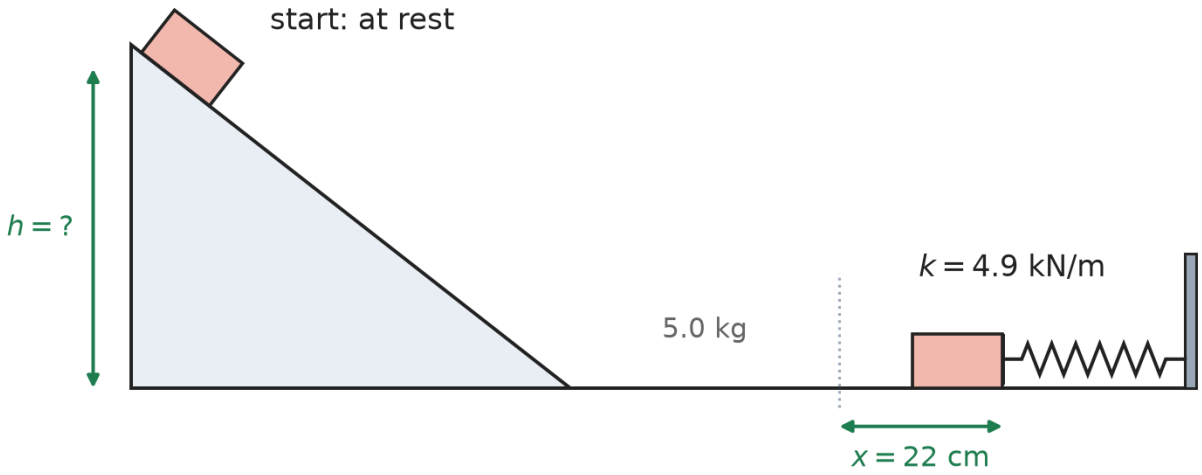
$x_{max} = 22$ cm $= 0.22$ m

$x_2 = 15$ cm $= 0.15$ m

$h = ?$

$v = ?$

The car at release and at maximum compression (h = 0 on the horizontal surface)



(a) The spring experiences a maximum compression of 22 cm. Determine the height of the initial release point. Include a diagram.

Start: at rest at height h, so the energy is all gravitational. End: at rest against the fully compressed spring, so it is all elastic. No friction anywhere.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mgh = \frac{1}{2}kx_{max}^2$

Divide both sides by mg: $h = \frac{kx_{max}^2}{2mg}$

Put in the numbers: $h = \frac{(4900)(0.22)^2}{2(5.0)(9.81)} = \frac{237.16}{98.1} = 2.4175 \text{ m}$

(b) Calculate the speed of the model car when the spring has been compressed 15 cm.

Compare this moment with the moment of maximum compression. Both have the same total energy. At 15 cm the car is still moving, so it has kinetic and elastic energy.

Formula: $E_{TOT2} = E_{TOTf}$

Put in the energies: $\frac{1}{2}mv^2 + \frac{1}{2}kx_2^2 = \frac{1}{2}kx_{max}^2$

Subtract $\frac{1}{2}kx_2^2$, multiply by $\frac{2}{m}$: $v^2 = \frac{k\left(x_{max}^2 - x_2^2\right)}{m}$

Put in the numbers: $v = \sqrt{\frac{(4900)\left((0.22)^2 - (0.15)^2\right)}{5.0}}$

$v = \sqrt{\frac{(4900)(0.0259)}{5.0}} = 5.038 \text{ m/s}$

(c) Determine the maximum acceleration of the car after it hits the spring.

The spring force is the only horizontal force on the car, and it is largest at maximum compression. It pushes the car back, away from the spring.

Formula: $F_{net} = ma$

Put in $F_{net} = F_e = kx_{max}$: $kx_{max} = ma$

Divide both sides by m: $a = \dfrac{kx_{max}}{m}$

Put in the numbers: $a = \dfrac{(4900)(0.22)}{5.0} = 215.6 \text{ m/s}^2$

FINAL ANSWER

(a) h = 2.4 m

(b) v = 5.0 m/s

(c) $\vec{a}$ = 2.2 × 10^2 m/s^2 [away from the spring]